One-Step Equations

AQA · KS3 Year 8

Name: ____________________ Class: __________ Date: __________

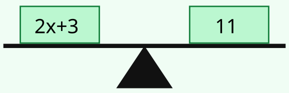
Solve a linear equation that needs a single step. Use the inverse operation to undo what has been done to the unknown, doing the same to both sides to keep the equation balanced.

VISUAL EXAMPLE

$2x + 3 = 11$

-3 both sides $2x = 8$

÷2 both sides $x = 4$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

Only changing one side — whatever you do to one side you must do to the other.

Using the wrong inverse: the inverse of $\times 3$ is $\div 3$, not $- 3$.

For $x \div 4 = 3$ you multiply both sides by 4; do not divide.

Sign slips with negatives — for example, $x + 7 = 2$ gives $x = -5$, not 5.

METHOD

1. Look at what has been done to the unknown letter.
2. Choose the inverse operation that undoes it.
3. Apply that inverse to BOTH sides of the equation.
4. Simplify each side to get the letter on its own.
5. Check by substituting your answer back into the original equation.

WORKED EXAMPLE

Solve: $x + 7 = 12$

Step 1 — 7 has been added to x.

Step 2 — The inverse of + 7 is - 7.

Step 3 — $x + 7 - 7 = 12 - 7$

Step 4 — $x = 5$

Step 5 — Check: $5 + 7 = 12$

1 Solve. (2 marks)

a) $x + 5 = 12$
b) $y - 3 = 10$
c) $m + 8 = 8$

HOW TO START

A number has been added to the letter — undo it by subtracting the same number from both sides.
For $x + 5 = 12$, subtract 5 from each side: $x = 12 - 5$.

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

2 Solve. (2 marks)

a) $4x = 20$
b) $x \div 3 = 6$
c) $7x = 56$

HOW TO START

Here the letter has been multiplied or divided.
For $4x = 20$, the inverse of $\times 4$ is $\div 4$: $x = 20 \div 4$.
For $x \div 3 = 6$, the inverse of $\div 3$ is $\times 3$: $x = 6 \times 3$.

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

3 Solve. (2 marks)

a) $x - 9 = -2$

b) $x + 11 = 4$

c) $6x = -30$

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

4 Solve. (3 marks)

a) $x \div 5 = -4$

b) $-3x = 21$

c) $x + 2.5 = 6$

ANSWERS

(a) = ____________________

(b) = ____________________

(c) = ____________________

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

5 Solve. The letter is on the right this time. (3 marks)

a) $9 = x + 4$

b) $20 = 5x$

c) $-8 = x - 3$

ANSWERS

(a) = ______

(b) = ______

(c) = ______

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

6 Form an equation and solve it. (3 marks)

a) A number n is multiplied by 4 to give 28. Find n.

b) When 5 is subtracted from a number p, the result is 12. Find p.

ANSWERS

(a) = ____________________

(b) = ____________________

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

7 A rectangle has width w cm and length 5w cm. (3 marks)

The length is 35 cm.

a) Write an equation in w. (1 mark)

b) Solve it to find the width w. (1 mark)

c) Work out the perimeter of the rectangle. (1 mark)

ANSWERS

(a) = ______________________

(b) = ______________________

(c) = ______________________

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

8 A delivery company charges the same fixed price £x for each parcel. (5 marks)
Four parcels cost £26 in total.

a) Write an equation for the total cost. (1 mark)
b) Solve it to find the price of one parcel. (2 marks)
c) A customer has a £40 budget. What is the greatest number of whole parcels they can send? Show your working. (2 marks)

ANSWERS

(a) = ______________________

(b) = ______________________

(c) = ______________________

METHOD 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

Self-reflection

How confident do you feel with this topic now? (tick one)

- [ ] Not yet
- [ ] Getting there
- [ ] Confident
- [ ] Really confident

Which question did you find the hardest, and why?

How do I decide which inverse operation to use?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $x = 7$ b) $y = 13$ c) $m = 0$

Q2: a) $x = 5$ b) $x = 18$ c) $x = 8$

Q3: a) $x = 7$ b) $x = -7$ c) $x = -5$

Q4: a) $x = -20$ b) $x = -7$ c) $x = 3.5$

Q5: a) $x = 5$ b) $x = 4$ c) $x = -5$

Q6: a) $4n = 28$ $n = 7$ b) $p - 5 = 12$ $p = 17$

Q7: a) $5w = 35$ b) $w = 7$ cm c) Perimeter = $2(7 + 35) = 84$ cm

Q8: a) $4x = 26$ b) $x = £6.50$ c) $40 \div 6.50 = 6.15...$ 6 whole parcels ($6 \times £6.50 = £39$ $£40$)